The tempered-likelihood learner: main result

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Data. Fix $d \geq 1$, a compact set $X \subseteq [-1,1]^d$ and any probability distribution D on $X \times [-1,1]$: the labels may be noisy or an arbitrary function of the input, but they lie in $[-1,1]$. The learner sees n independent samples $S = ((x_i, y_i))_{i=1}^n$ from D. The test loss of $f : X \to \mathbb{R}$ is $L_D(f) = \mathbb{E}_{(x,y)\sim D}\,(f(x) - y)^2$.

Networks. A network of depth $L \geq 2$ and width N has parameters

$$w = (W^1, W^2, \dots, W^{L-1}, v) \in \mathbb{R}^{N\times(d+1)} \times \left(\mathbb{R}^{N\times N}\right)^{L-2} \times \mathbb{R}^N$$

and computes

$$z_w(x) = v^\top \tanh\bigl(W^{L-1} \cdots \tanh\bigl(W^1 (1, x)\bigr) \cdots\bigr),$$

with tanh applied entrywise; biases enter only through the constant input coordinate. Predictions are clipped to the label range, $\operatorname{clip}(u) = \max\{-1, \min\{1, u\}\}$.

The learner. Fix an exponent $\theta \in (0,1)$, a width N and a scale $\sigma > 0$. The prior makes all weights independent centred Gaussians, with variance $N\sigma^2/d$ in W^1, σ^2 in $W^2, \dots, W^{L-1}$ and σ^2/N in v. The likelihood is

$$y_i \mid x_i, w \;\sim\; \mathrm{N}\bigl(z_w(x_i),\, n^{1-\theta}\sigma^2\bigr),$$

Gaussian noise whose variance is tempered by the factor $n^{1-\theta}$. Let Π_σ be the posterior given S, and take $\sigma \downarrow 0$ with n, S and N fixed. The learner's test loss is that of a clipped posterior draw, $\mathbb{E}_{w\sim\Pi_\sigma} L_D(\operatorname{clip} z_w)$.

Theorem. Fix the depth $L \geq 2$. Below, $\mathrm{poly}(\dots)$ is a fixed polynomial in the quantities listed; nothing else enters it. For every d, X, D, $n \geq 1$ and $\delta \in (0,1)$:

a. *Against one network.* Take $\theta = 1/(L+1)$. For every network g of depth L, width m and weights in $[-B, B]$, with probability at least $1-\delta$ over S, at every learner width $N \geq m$,

$$\limsup_{\sigma\downarrow 0}\; \mathbb{E}_{w\sim\Pi_\sigma} L_D(\operatorname{clip} z_w) \;\leq\; L_D(g) + \mathrm{poly}\bigl(L, d, m, B, \log(1/\delta)\bigr) \cdot \sqrt{\log(n+1)} \cdot n^{-\frac{1}{L+1}}.$$

b. *Against all networks of size n^α.* Let $L \geq 3$ and $0 < \alpha < \frac{1}{4(L-1)}$, and take $\theta = \frac{1+8\alpha}{L+1}$. With probability at least $1-\delta$ over S, at every learner width $N \geq n^\alpha$,

$$\limsup_{\sigma\downarrow 0}\; \mathbb{E}_{w\sim\Pi_\sigma} L_D(\operatorname{clip} z_w) \;\leq\; \min_f L_D(f) + \mathrm{poly}\bigl(L, d, \log(1/\delta)\bigr) \cdot \sqrt{\log(n+1)} \cdot n^{-\frac{1-4\alpha(L-1)}{L+1}},$$

where f ranges over the networks of depth L with width at most n^α and weights in $[-n^\alpha, n^\alpha]$.

Both bounds also hold for the posterior mean of the clipped outputs, $x \mapsto \mathbb{E}_{w\sim\Pi_\sigma} \operatorname{clip} z_w(x)$.

Source and status. Parts (a) and (b) are Theorem 6.3 and Corollary 8.1 of `research-notes/B-tempered-likelihood.md`, with the error terms bounded by explicit polynomials as in Corollary 3 of `statement/tempered-learner.pdf`; both files are in `neural-network-generalization/works-in-progress/agnostic-learning/`. The proofs in B are paper proofs refereed by independent AI agents, not Lean-verified. The polynomial bounds were checked by independent AI agents but not refereed.

Limits. Any $\theta \in (0, \frac{1}{L-1})$ works in (a), with rate $n^{-\min\{\theta,\,(1-\theta(L-1))/2\}}$ in place of $n^{-1/(L+1)}$. For $L \geq 3$ the range is sharp in the worst case: with $\theta > \frac{1}{L-1}$ the learner can memorize label noise (B, Theorem 7.4), and without tempering (variance σ^2 instead of $n^{1-\theta}\sigma^2$) it can do worse than the zero network on pure-noise labels (A, Theorem 4.4). Bounded competitor weights are needed: for $L \geq 3$ and $\alpha > \frac{L}{2(L-1)(2L-1)}$, no tempering gives a bound like (b) that is polynomial, uniform over D and d, and valid at every learner width $N \geq n^\alpha$ (B, Corollary 9.3).

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